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Research Interests: Industrial Organization, Generative AI, Microeconomics
Creative Works

EDUCATION

Ph.D. in Economics, University of Toronto	2026 (Expected)
<i>Committee:</i> Matthew Mitchell (co-supervisor), Heski Bar-Isaac (co-supervisor) Avi Goldfarb, Ambarish Chandra	
M.A in Economics, University of Toronto	2020
B.A in Economics, Fudan University	2019

RESEARCH

Evolving With GenAI (JMP)**Hiding From Generative AI**, R&R at Quantitative Economics**Platforms in Platform**

AWARDS AND GRANTS

CREATE for Accelerated Discovery (AccelD) Stipend Award, University of Toronto	2025
TD MDAL Grant, Rotman School of Management, University of Toronto	2025
TD MDAL Grant, Rotman School of Management, University of Toronto	2024
University of Toronto Doctoral Fellowship	2020 - 2025

CONFERENCE PRESENTATIONS

TSE Digital Economics Conference (Toulouse)	2025
North American Summer Meeting of Econometric Society (Nashville)	2024
ESIF Economics and AI+ML Meeting of Econometric Society (Ithaca)	2024
European Economic Association Annual Meeting (Rotterdam)	2024
American Law and Economics Annual Meeting (Ann Arbor)	2024
Canadian Economics Association Annual Meeting (Toronto)	2024

COMPUTER SKILLS

Python, STATA, MATLAB, LaTeX, Cloud Computing

REFERENCES

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Abstracts

Evolving With GenAI

(Job Market Paper)

What is the role of human input when producing with generative AI? In the context of language-based AI, human input refers to prompts, which are combinations of words that humans use to interact with AI. What distinguishes human input from purely automated AI processes is *adaptation*: users dynamically modify their prompts in response to AI output. In this paper, I focus on two types of adaptation: (1) adaptation to new AI versions, referring to how people change their prompts in response to AI upgrades; (2) adaptation to outputs from previous prompts, referring to how people adjust their prompts iteratively and eventually converge to their desired outcomes. To explore the role of these two types of adaptation, I study how users interact with Midjourney, a popular AI image generator, by analyzing submitted prompts and generated images. Firstly, I examine the role of adaptation to different AI versions by analyzing prompts across four Midjourney versions during the data observation period and show that word frequency changes. By submitting prompts written for the old version to the new one and vice versa, I decompose output shifts and show that 73% of output shifts result from changing prompts alone, 20% from changing AI alone, and 7% from AI-human cross effect. These numbers imply that AI and human input are complements. Secondly, I study the role of adaptation to previous output by analyzing how prompts evolve in a session. I estimate a structural model of creative process using the framework of Weitzman (1979) and Hodgson and Lewis (2025) in which users judge image quality from the preceding prompt and change the next prompt. In counterfactual analysis, I show that when users are not allowed to judge image quality and adjust prompts accordingly, 313% more prompts are needed in each session to reach the same final prompt as observed in data.

Hiding From Generative AI

How does generative Artificial Intelligence (AI) change the behaviors of content creators? I investigate the effect of an AI image generator on artists' incentives to publish artworks using data from an online art platform, DeviantArt. On November 11 2022, DeviantArt introduced a generative AI image generator into the platform and artworks on this platform entered training data by default. Using a difference-in-differences estimation with artists who do not use AI, I show that digital artists publish 21% fewer artworks following AI's introduction on this platform, in contrast to artisan crafts artists. This reduction could potentially hinder knowledge spillovers to other artists and AI training data availability. By matching the artworks of artists who publish both on DeviantArt and Instagram, I find that despite artists publishing fewer artworks on DeviantArt, the quality of published artworks for a given artist remains the same after the introduction of AI.

Platforms in Platform

Platforms could be part of a larger platform like an app store. Even though there exists a positive cross-group network externality between buyers and sellers, there is still an incentive for buyers and sellers to segment into different platforms. Buyers may choose to use different platforms to access more targeted products, while sellers may use different platforms to avoid competition for buyers' attention. This paper argues that segmenting the market using smaller platforms can be profitable for the large platform, as it can charge different platform fees to sellers of different quality and save buyers' attention to better sellers. This paper also shows that when market agglomeration is more profitable for the large platform, it may choose to ban smaller platforms, which could be socially inefficient as it may deter mediocre sellers from entering the market.